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JEE Main Advanced Practice Sheet — Physics

Topic: Applications of Conservation of Momentum (Recoil of Gun, Variable Mass & Rocket Propulsion)
Time Allowed: 60 Minutes — **Max Marks:** 80 — **Marking Scheme:** +4, -1

Q1. A rifle of mass 4 kg fires a bullet of mass 40 g with a muzzle velocity of 800 m/s relative to the barrel. If the rifle is free to move, the recoil velocity of the rifle is:

(A) 8.0 m/s
(B) 7.92 m/s
(C) 8.08 m/s
(D) 4.0 m/s

Q2. A shell of mass M at rest is fired from a gun of mass M_g. If the ratio of mass of the gun to that of the shell is n ($M_g/M = n$), the ratio of the kinetic energy of the shell to that of the recoiling gun is:

(A) n
(B) $1/n$
(C) n^2
(D) $\sqrt{n}$

Q3. A machine gun fires 360 bullets per minute, each of mass 20 g with a muzzle speed of 600 m/s. The average force that the gunner must exert on the gun to hold it in place is:

(A) 36 N
(B) 72 N
(C) 144 N
(D) 432 N

Q4. A cannon of mass M mounted on a smooth horizontal surface fires a shell of mass m with a velocity v_0 at an angle θ relative to the horizontal barrel. The recoil speed of the cannon along the horizontal surface is:

(A) $\dfrac{mv_0\cos\theta}{M}$
(B) $\dfrac{mv_0\cos\theta}{M+m}$
(C) $\dfrac{mv_0\cos\theta}{M-m\cos\theta}$
(D) $\dfrac{mv_0}{M+m\cos\theta}$

Q5. A man of mass 60 kg stands on a trolley of mass 140 kg on smooth horizontal rails. The system is initially at rest. If the man jumps horizontally with a velocity of 4 m/s relative to the trolley, the absolute recoil speed of the trolley is:

(A) 1.2 m/s
(B) 1.71 m/s
(C) 2.0 m/s
(D) 0.8 m/s

Q6. A gun of mass 100 kg fires a projectile of mass 1 kg with a speed of 400 m/s. The gun recoils against a spring of spring constant $k = 4\times10^4$ N/m. The maximum compression of the spring is:

(A) 10 cm
(B) 20 cm
(C) 40 cm
(D) 5 cm

Q7. A gun of mass M fires a bullet of mass m horizontally with speed v. The coefficient of friction between the gun and the horizontal ground is μ. The distance travelled by the gun before coming to rest is:

(A) $\dfrac{m^2v^2}{2\mu M^2 g}$
(B) $\dfrac{mv^2}{2\mu M g}$
(C) $\dfrac{M^2v^2}{2\mu m^2 g}$
(D) $\dfrac{m^2v^2}{\mu M^2 g}$

Q8. A rocket of initial mass $M_0 = 5000$ kg is fired vertically upwards. The exhaust velocity of gases relative to the rocket is $u = 1200$ m/s. If the rocket is to have an initial upward acceleration of 20 m/s^2, the rate of consumption of fuel must be ($g = 10$ m/s^2):

(A) 125 kg/s
(B) 83.3 kg/s
(C) 41.7 kg/s
(D) 150 kg/s

Q9. A rocket of mass 6000 kg is set for vertical firing. If the exhaust speed of its fuel is 1000 m/s, the minimum rate of ejection of gas required just to overcome the weight of the rocket and hover is ($g = 9.8$ m/s^2):

(A) 58.8 kg/s
(B) 60.0 kg/s
(C) 30.0 kg/s
(D) 117.6 kg/s

Q10. In a gravity-free space, a rocket consumes fuel at a constant rate and ejects it with a relative velocity u. If the ratio of initial mass to final mass at burnout is e^3, the final velocity acquired by the rocket starting from rest is:

(A) u
(B) $2u$
(C) $3u$
(D) e^3u

Q11. Sand drops vertically at a steady rate of $\mu = 5$ kg/s onto a horizontal conveyor belt moving uniformly with a speed of $v = 2$ m/s. The additional power required from the motor to keep the belt moving at the same constant speed is:

(A) 10 W
(B) 20 W
(C) 40 W
(D) 5 W

Q12. An open cart of mass M_0 is moving horizontally with velocity v_0 on a frictionless surface. Sand starts falling vertically into the cart at a constant rate λ kg/s. The velocity of the cart as a function of time t is:

(A) v_0
(B) $\dfrac{M_0 v_0}{M_0 + \lambda t}$
(C) $v_0 - \dfrac{\lambda t}{M_0}$
(D) $v_0 e^{-\lambda t / M_0}$

Q13. A cart of mass M filled with sand of mass m_s is moving with velocity v_0 on a frictionless track. Sand leaks vertically downwards through a small hole at the bottom at a constant rate $r = \frac{dm}{dt}$. The speed of the cart after all the sand has leaked out is:

(A) $v_0 \left(\dfrac{M + m_s}{M} \right)$
(B) v_0
(C) $v_0 \left(\dfrac{M}{M + m_s} \right)$
(D) $v_0 \ln \left(\dfrac{M + m_s}{M} \right)$

Q14. A projectile of mass $3m$ is fired with an initial speed u at an angle θ with the horizontal. At the highest point of its trajectory, it explodes into two fragments of mass m and $2m$. If the lighter fragment (m) falls vertically downwards immediately after explosion, the horizontal distance from the point of projection where the heavier fragment ($2m$) lands is:

(A) $\dfrac{3}{2}R$
(B) $2R$
(C) $\dfrac{7}{4}R$
(D) $\dfrac{5}{4}R$

(where $R = \dfrac{u^2 \sin 2\theta}{g}$ *is the original range)*

Q15. A flatcar of mass M is initially at rest on a frictionless horizontal track. A man of mass m runs from one end of the car to the other (length L) with a speed u relative to the flatcar. The displacement of the flatcar relative to the ground during this process is:

(A) $\dfrac{mL}{M}$
(B) $\dfrac{mL}{M + m}$
(C) $\dfrac{ML}{M + m}$
(D) $\dfrac{mL}{M - m}$

Q16. A rifle on a frictionless horizontal table fires n identical bullets of mass m consecutively. The velocity of each bullet relative to the rifle barrel at the moment of firing is u. If the mass of the rifle without bullets is M, the recoil velocity of the rifle after all n bullets are fired is:

(A) $u \sum_{k=1}^{n} \dfrac{m}{M + km}$
(B) $u \sum_{k=1}^{n} \dfrac{m}{M + (k-1)m}$
(C) $\dfrac{nmu}{M + nm}$
(D) $\dfrac{nmu}{M}$

Q17. A jet of water from a nozzle of cross-sectional area A strikes a stationary cart of mass M horizontally with a speed v_0 and bounces off perpendicularly. Assuming no resistance, the acceleration of the cart when its speed becomes v is (ρ = density of water):

(A) $\dfrac{\rho A v_0^2}{M}$
(B) $\dfrac{\rho A (v_0 - v)^2}{M}$
(C) $\dfrac{\rho A (v_0^2 - v^2)}{M}$
(D) $\dfrac{2\rho A (v_0 - v)^2}{M}$

Q18. In a rocket, to maintain a constant acceleration $a = 2g$ upwards in a uniform gravitational field g where relative exhaust speed is u, the mass of the rocket must decrease with time according to the relation:

(A) $M(t) = M_0 e^{-3gt/u}$
(B) $M(t) = M_0 e^{-2gt/u}$
(C) $M(t) = M_0 (1 - 3gt/u)$
(D) $M(t) = M_0 e^{-gt/u}$

Q19. A bullet of mass m moving with horizontal velocity v strikes and gets embedded in a wooden block of mass M suspended by a light string of length l. The minimum value of v so that the block completes a full vertical circle is:

(A) $\left(\dfrac{M + m}{m} \right) \sqrt{5gl}$
(B) $\left(\dfrac{m}{M + m} \right) \sqrt{5gl}$
(C) $\left(\dfrac{M}{m} \right) \sqrt{5gl}$
(D) $\sqrt{5gl}$

Q20. A 2-stage rocket has total initial mass M_0. The first stage has fuel mass $0.8M_0$ and structural mass $0.1M_0$. The second stage has fuel mass $0.08M_0$ and payload mass $0.02M_0$. The exhaust speed for both stages is u. In gravity-free space, the final burnout speed of the payload starting from rest (discarding first stage structure before firing second stage) is:

(A) $u(\ln 5 + \ln 5) = 2u \ln 5$

(B) $u(\ln 10 + \ln 5)$

(C) $u(\ln 5 + \ln 10)$

(D) $u \ln 50$

SMAPhysics.com — Answer Key & Detailed Solutions

Applications of Conservation of Momentum — Recoil & Rocket Propulsion

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	B	6	B	11	B	16	A
2	A	7	A	12	B	17	B
3	B	8	A	13	B	18	A
4	B	9	A	14	C	19	A
5	A	10	C	15	B	20	A

Detailed Step-by-Step Solutions

Sol 1. Option (B)
Let $M = 4$ kg, $m = 40$ g $= 0.04$ kg. The given speed $v_{rel} = 800$ m/s is relative to the barrel.
If the rifle recoils with speed V backwards relative to ground, the speed of the bullet relative to ground is $v_b = v_{rel} - V$.
By conservation of linear momentum:

$$0 = m(v_{rel} - V) - MV \implies (M+m)V = mv_{rel}$$

$$V = \frac{mv_{rel}}{M+m} = \frac{0.04 \times 800}{4 + 0.04} = \frac{32}{4.04} \approx 7.92 \text{ m/s}$$

Sol 2. Option (A)
From conservation of momentum:

$$M_g V_g = M v_s \implies V_g = \frac{M v_s}{M_g} = \frac{v_s}{n}$$

Ratio of kinetic energies:

$$\frac{K_{shell}}{K_{gun}} = \frac{\frac{1}{2}M v_s^2}{\frac{1}{2}M_g V_g^2} = \frac{M v_s^2}{M_g \left(\frac{v_s}{n}\right)^2} = \frac{M n^2}{M_g} = \frac{n^2}{n} = n$$

Sol 3. Option (B)
Number of bullets per second $N = \frac{360}{60} = 6 \text{ s}^{-1}$.
Mass of each bullet $m = 20$ g $= 0.02$ kg, speed $v = 600$ m/s.
Rate of change of momentum is the required holding force:

$$F_{avg} = N \cdot m \cdot v = 6 \times 0.02 \times 600 = 72 \text{ N}$$

Sol 4. Option (A)
Relative to the barrel, the projectile's horizontal velocity component is $v_0 \cos\theta$.
If cannon recoils with horizontal speed V backwards relative to ground, horizontal speed of shell relative to ground is $v_{sx} = v_0 \cos\theta - V$.
Since no external horizontal force acts:

$$m(v_0 \cos\theta - V) = MV \implies V = \frac{mv_0 \cos\theta}{M+m}$$

Note: If the muzzle velocity v_0 is specified relative to the gun barrel, $V = \frac{mv_0 \cos\theta}{M+m}$, which corresponds to Option (B). If v_0 were ground speed, it would be $\frac{mv_0\cos\theta}{M}$. Hence Option (B) is the exact answer.

Sol 5. Option (A)
Let trolley recoil with speed V to the left.
Speed of man relative to ground $= (u_{rel} - V) = 4 - V$.
By conservation of momentum in the horizontal direction:

$$m(4 - V) - MV = 0 \implies (M+m)V = m(4)$$

$$V = \frac{60 \times 4}{140 + 60} = \frac{240}{200} = 1.2 \text{ m/s}$$

Sol 6. Option (B)
Recoil velocity of the gun V:

$$MV = mv \implies V = \frac{mv}{M} = \frac{1 \times 400}{100} = 4 \text{ m/s}$$

By conservation of mechanical energy for the recoiling gun and spring:

$$\frac{1}{2}MV^2 = \frac{1}{2}kx_{max}^2$$

$$x_{max} = \sqrt{\frac{MV^2}{k}} = \sqrt{\frac{100 \times 4^2}{4 \times 10^4}} = \sqrt{\frac{1600}{40000}} = \sqrt{0.04} = 0.2 \text{ m} = 20$$

Sol 7. Option (A)
Recoil velocity of gun $V = \frac{mv}{M}$.
Deceleration of gun due to friction: $a = \mu g$.
Using $V^2 = 2as$:

$$s = \frac{V^2}{2\mu g} = \frac{(mv/M)^2}{2\mu g} = \frac{m^2 v^2}{2\mu M^2 g}$$

Sol 8. Option (A)
Equation of motion of the rocket:

$$F_{thrust} - M_0 g = M_0 a \implies u\left(\frac{dm}{dt}\right) = M_0(g + a)$$

Given $M_0 = 5000$ kg, $u = 1200$ m/s, $a = 20 \text{ m/s}^2$, $g = 10 \text{ m/s}^2$:

$$\frac{dm}{dt} = \frac{M_0(g+a)}{u} = \frac{5000(10+20)}{1200} = \frac{5000 \times 30}{1200} = 125 \text{ kg/s}$$

Sol 9. Option (A)
To just hover/overcome weight:

$$F_{thrust} = Mg \implies u\left(\frac{dm}{dt}\right) = Mg$$

$$\frac{dm}{dt} = \frac{Mg}{u} = \frac{6000 \times 9.8}{1000} = 58.8 \text{ kg/s}$$

Sol 10. Option (C)
Tsiolkovsky rocket equation in gravity-free space:

$$v = u \ln\left(\frac{M_0}{M}\right)$$

Given $\frac{M_0}{M} = e^3$:

$$v = u\ln(e^3) = 3u$$

Sol 11. Option (B)
The incoming sand has zero initial horizontal velocity. Rate of momentum imparted to the sand:

$$F_{ext} = v\frac{dm}{dt} = 2 \times 5 = 10 \text{ N}$$

Power required by the motor $P = F_{ext} \cdot v$:

$$P = 10 \text{ N} \times 2 \text{ m/s} = 20 \text{ W}$$

Note: Half of the power ($\frac{1}{2}\mu v^2 = 10$ W) increases kinetic energy of sand, and the other half is dissipated as heat during impact.

Sol 12. Option (B)
No external horizontal force acts on the system (cart + sand).
Total initial momentum $= M_0 v_0$.
At time t, total mass is $M(t) = M_0 + \lambda t$.
Since falling sand has zero horizontal momentum before landing:

$$(M_0 + \lambda t)v(t) = M_0 v_0 \implies v(t) = \frac{M_0 v_0}{M_0 + \lambda t}$$

Sol 13. Option (B)
When sand leaks vertically out of a hole, each grain leaves the cart carrying the horizontal velocity of the cart at that instant.
Therefore, the leaking mass carries away its own momentum and exerts zero reaction force on the cart:

$$F_{thrust} = v_{rel}\frac{dm}{dt} = 0 \quad (\because v_{rel} = 0)$$

Thus, the acceleration is zero and the speed of the cart remains constant at v_0.

Sol 14. Option (C)
At highest point, velocity is horizontal: $v_x = u\cos\theta$.
Total horizontal momentum before explosion:

$$P_x = (3m)u\cos\theta$$

After explosion, the piece m drops vertically, so its horizontal velocity is 0.
Let v' be horizontal velocity of piece $2m$:

$$(2m)v' = (3m)u\cos\theta \implies v' = \frac{3}{2}u\cos\theta$$

Time to fall from highest point is $t = \frac{T}{2}$.
Horizontal distance covered from explosion point:

$$x_2 = v' \cdot \frac{T}{2} = \left(\frac{3}{2}u\cos\theta\right)\left(\frac{u\sin\theta}{g}\right) = \frac{3}{4}R$$

Total distance from projection point $= \frac{R}{2} + \frac{3}{4}R = \frac{7}{4}R$.

Sol 15. Option (B)
Since no external horizontal force acts, the center of mass of (man + car) remains stationary:

$$\Delta x_{cm} = 0 \implies m\Delta x_{man} + M\Delta x_{car} = 0$$

Let displacement of car be x in opposite direction:

$$m(L - x) = Mx \implies x = \frac{mL}{M+m}$$

Sol 16. Option (A)
For the k-th bullet, mass of remaining rifle system before shot is $M_k = M + km$.
The increment in rifle velocity when k-th bullet is fired is:

$$\Delta V_k = \frac{m}{M+km}u$$

Total recoil velocity after n shots:

$$V = \sum_{k=1}^{n} \Delta V_k = u\sum_{k=1}^{n}\frac{m}{M+km}$$

Sol 17. Option (B)
Relative velocity of water striking the cart moving at speed v is $(v_0 - v)$.
Mass rate of water striking the cart:

$$\frac{dm}{dt} = \rho A(v_0 - v)$$

Force exerted on the cart:

$$F = \frac{dm}{dt}(v_0 - v) = \rho A(v_0 - v)^2$$

Acceleration $a = \frac{F}{M} = \frac{\rho A(v_0 - v)^2}{M}$.

Sol 18. Option (A)
Equation of motion:

$$-u\frac{dM}{dt} - Mg = Ma$$

Given $a = 2g$:

$$-u\frac{dM}{dt} = 3Mg \implies \frac{dM}{M} = -\frac{3g}{u}dt$$

Integrating both sides from M_0 to $M(t)$:

$$\ln\left(\frac{M(t)}{M_0}\right) = -\frac{3gt}{u} \implies M(t) = M_0 e^{-3gt/u}$$

Sol 19. Option (A)
By conservation of momentum during collision:

$$mv = (M+m)V \implies V = \frac{mv}{M+m}$$

For the block to complete vertical circular motion, velocity at bottom must be at least $\sqrt{5gl}$:

$$V \geq \sqrt{5gl} \implies \frac{mv}{M+m} \geq \sqrt{5gl}$$

$$v_{min} = \left(\frac{M+m}{m}\right)\sqrt{5gl}$$

Sol 20. Option (A)
Stage 1: Initial mass $M_{01} = M_0$.
Mass after Stage 1 fuel burn: $M_{f1} = 0.1M_0 + 0.08M_0 + 0.02M_0 = 0.2M_0$.
Velocity gain in Stage 1:

$$\Delta v_1 = u\ln\left(\frac{M_0}{0.2M_0}\right) = u\ln 5$$

Stage 1 structure ($0.1M_0$) is discarded.
Stage 2 initial mass: $M_{02} = 0.08M_0 + 0.02M_0 = 0.10M_0$.
Stage 2 final mass (payload only): $M_{f2} = 0.02M_0$.
Velocity gain in Stage 2:

$$\Delta v_2 = u\ln\left(\frac{0.10M_0}{0.02M_0}\right) = u\ln 5$$

Total burnout velocity:

$$v_{total} = \Delta v_1 + \Delta v_2 = u\ln 5 + u\ln 5 = 2u\ln 5$$